

Practice Problems:

1. Ten cards numbered 1 to 10 are placed in a box, mixed up thoroughly and then one card is drawn randomly. If it is known that the number on the drawn card is more than 3, what is the probability that it is an even number?
2. A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?
3. Compute $P(E|F)$ in the following,
Mother, father, and son line up at random for a family picture
 E : son on one end,
 F : father in middle

4. Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls given that
- (i) the youngest is a girl,
 - (ii) at least one is a girl?
5. Given that the two numbers appearing on throwing two dice are different. Find the probability of the event 'the sum of numbers on the dice is 4.'
6. An urn contains 10 black & 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black?
7. A die is thrown. If E is the event 'the number appearing is a multiple of 3' and F be the event 'the number appearing is even' then find whether E and F are independent?
8. Three coins are tossed simultaneously. Consider the event E 'three heads' or

three tails', F 'at least two heads' and G 'at most two heads'. Of the pairs (E, F) , (E, G) and (F, G) , which are independent? which are dependent?

9. If A and B are two events such that $\Pr(A) = 1/4$, $\Pr(B) = 1/2$ and $\Pr(A \cap B) = 1/8$, find $\Pr(A^c \cap B^c)$.
10. A die is tossed thrice. Find the probability of getting an odd number at least once.